

Chapter 1 MPLS VPN Configuration

Lab 1-1 MPLS LDP Configuration

Learning Objectives

The objectives of this lab are to learn and understand:

- How to enable and disable MPLS
- How to enable and disable MPLS LDP
- How to configure LSPs using MPLS LDP
- How to configure the LDP LSP trigger policy on an MPLS router

Topology

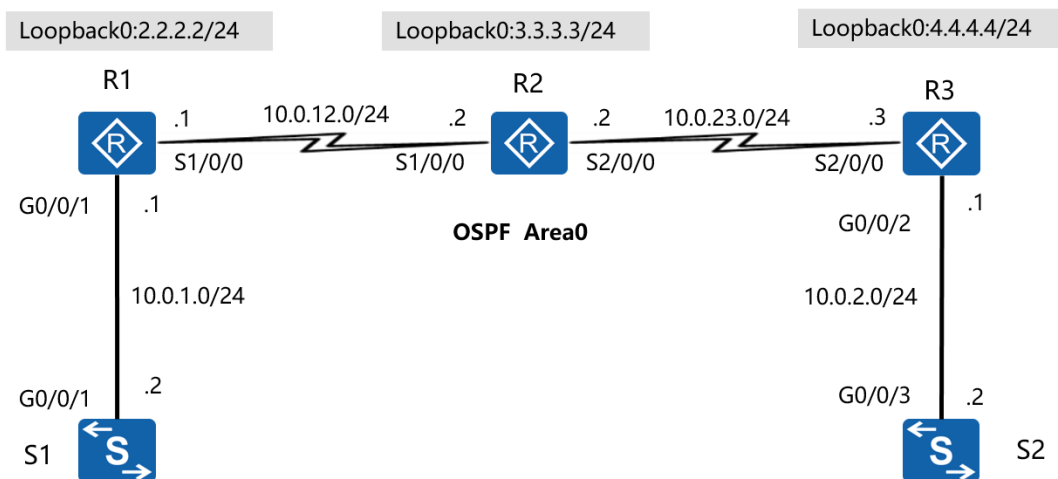


Figure 1-1 MPLS LDP topology

Scenario

Assume that you are a network administrator of an enterprise. Your enterprise uses an IP network with poor forwarding performance. You need to use MPLS to improve the forwarding rate of routers. Static LSPs are configured manually, while LDP is a protocol developed for label distribution. To perform flexible configuration, use LDP to set up MPLS LSPs.

Tasks

Step 1 **Perform basic configurations and configure IP addresses.**

Configure IP addresses and masks for all routers.

```
<Huawei>system-view
```

Enter system view, return user view with Ctrl+Z.

```
[Huawei]sysname S1
```

```
[S1]interface Vlanif 1
```

```
[S1-Vlanif1]ip address 10.0.1.2 24
```

```
<Huawei>system-view
```

Enter system view, return user view with Ctrl+Z.

```
[Huawei]sysname R1
```

```
[R1]interface GigabitEthernet 0/0/1
```

```
[R1-GigabitEthernet0/0/1]ip address 10.0.1.1 24
```

```
[R1-GigabitEthernet0/0/1]quit
```

```
[R1]interface Serial 1/0/0
```

```
[R1-Serial1/0/0]ip address 10.0.12.1 24
```

```
[R1-Serial1/0/0]quit
```

```
[R1]interface loopback 0
```

```
[R1-LoopBack0]ip address 2.2.2.2 24
```

```
<Huawei>system-view
```

Enter system view, return user view with Ctrl+Z.

```
[Huawei]sysname R2
```

```
[R2]interface Serial 1/0/0
```

```
[R2-Serial1/0/0]ip address 10.0.12.2 24
```

```
[R2-Serial1/0/0]quit
```

```
[R2]interface Serial 2/0/0
```

[R2-Serial2/0/0]ip address 10.0.23.2 24

[R2-Serial2/0/0]quit

[R2]interface loopback 0

[R2-LoopBack0]ip address 3.3.3.3 24

<Huawei>system-view

Enter system view, return user view with Ctrl+Z.

[Huawei]sysname R3

[R3]interface GigabitEthernet 0/0/2

[R3-GigabitEthernet0/0/2]ip address 10.0.2.1 24

[R3-GigabitEthernet0/0/2]quit

[R3]interface Serial 2/0/0

[R3-Serial2/0/0]ip address 10.0.23.3 24

[R3-Serial2/0/0]quit

[R3]interface loopback 0

[R3-LoopBack0]ip address 4.4.4.4 24

<Huawei>system-view

Enter system view, return user view with Ctrl+Z.

[Huawei]sysname S2

[S2]interface Vlanif 1

[S2-Vlanif1]ip address 10.0.2.2 24

After the configurations are complete, test the connectivity of direct links.

Step 2 **Configure a single OSPF area.**

Add 10.0.12.0/24, 10.0.23.0/24, 10.0.1.0/24, and 10.0.2.0/24 to OSPF area 0.

[S1]ospf 1 router-id 1.1.1.1

[S1-ospf-1]area 0

[S1-ospf-1-area-0.0.0.0]network 10.0.1.0 0.0.0.255

```
[R1]ospf 1 router-id 2.2.2.2
```

```
[R1-ospf-1]area 0
```

```
[R1-ospf-1-area-0.0.0.0]network 10.0.1.0 0.0.0.255
```

```
[R1-ospf-1-area-0.0.0.0]network 10.0.12.0 0.0.0.255
```

```
[R1-ospf-1-area-0.0.0.0]network 2.2.2.0 0.0.0.255
```

```
[R2]ospf 1 router-id 3.3.3.3
```

```
[R2-ospf-1]area 0
```

```
[R2-ospf-1-area-0.0.0.0]network 10.0.12.0 0.0.0.255
```

```
[R2-ospf-1-area-0.0.0.0]network 10.0.23.0 0.0.0.255
```

```
[R2-ospf-1-area-0.0.0.0]network 3.3.3.0 0.0.0.255
```

```
[R3]ospf 1 router-id 4.4.4.4
```

```
[R3-ospf-1]area 0
```

```
[R3-ospf-1-area-0.0.0.0]network 10.0.23.0 0.0.0.255
```

```
[R3-ospf-1-area-0.0.0.0]network 10.0.2.0 0.0.0.255
```

```
[R3-ospf-1-area-0.0.0.0]network 4.4.4.0 0.0.0.255
```

```
[S2]ospf 1 router-id 5.5.5.5
```

```
[S2-ospf-1]area 0
```

```
[S2-ospf-1-area-0.0.0.0]network 10.0.2.0 0.0.0.255
```

Check the routing table and test connectivity on the entire network.

```
[R2]ping 10.0.1.2
```

```
  PING 10.0.1.2: 56 data bytes, press CTRL_C to break
```

```
    Reply from 10.0.1.2: bytes=56 Sequence=1 ttl=253 time=36 ms
```

```
    Reply from 10.0.1.2: bytes=56 Sequence=2 ttl=253 time=31 ms
```

Reply from 10.0.1.2: bytes=56 Sequence=3 ttl=253 time=31 ms

Reply from 10.0.1.2: bytes=56 Sequence=4 ttl=253 time=31 ms

Reply from 10.0.1.2: bytes=56 Sequence=5 ttl=253 time=31 ms

--- 10.0.1.2 ping statistics ---

5 packet(s) transmitted

5 packet(s) received

0.00% packet loss

round-trip min/avg/max = 31/32/36 ms

[R2]ping 10.0.2.2

PING 10.0.2.2: 56 data bytes, press CTRL_C to break

Reply from 10.0.2.2: bytes=56 Sequence=1 ttl=253 time=38 ms

Reply from 10.0.2.2: bytes=56 Sequence=2 ttl=253 time=33 ms

Reply from 10.0.2.2: bytes=56 Sequence=3 ttl=253 time=33 ms

Reply from 10.0.2.2: bytes=56 Sequence=4 ttl=253 time=33 ms

Reply from 10.0.2.2: bytes=56 Sequence=5 ttl=253 time=33 ms

--- 10.0.2.2 ping statistics ---

5 packet(s) transmitted

5 packet(s) received

0.00% packet loss

round-trip min/avg/max = 33/34/38 ms

Run the **display ip routing-table** command to check the OSPF routing table.

[R2]display ip routing-table

Route Flags: R - relay, D - download to fib

Routing Tables: Public

Destinations : 19

Routes : 19

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
2.2.2.2/32	OSPF	10	1562	D	10.0.12.1	Serial1/0/0
3.3.3.0/24	Direct	0	0	D	3.3.3.3	LoopBack0
3.3.3.3/32	Direct	0	0	D	127.0.0.1	InLoopBack0
3.3.3.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
4.4.4.4/32	OSPF	10	1562	D	10.0.23.3	Serial2/0/0
10.0.1.0/24	OSPF	10	1563	D	10.0.12.1	Serial1/0/0
10.0.2.0/24	OSPF	10	1563	D	10.0.23.3	Serial2/0/0
10.0.12.0/24	Direct	0	0	D	10.0.12.2	Serial1/0/0
10.0.12.1/32	Direct	0	0	D	10.0.12.1	Serial1/0/0
10.0.12.2/32	Direct	0	0	D	127.0.0.1	InLoopBack0
10.0.12.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
10.0.23.0/24	Direct	0	0	D	10.0.23.2	Serial2/0/0
10.0.23.2/32	Direct	0	0	D	127.0.0.1	InLoopBack0
10.0.23.3/32	Direct	0	0	D	10.0.23.3	Serial2/0/0
10.0.23.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

Step 3 Configure MPLS LDP.

Configure MPLS and LDP globally on MPLS routers.

[R1]mpls lsr-id 2.2.2.2

[R1]mpls

Info: Mpls starting, please wait... OK!

[R1-mpls]mpls ldp

[R2]mpls lsr-id 3.3.3.3

[R2]mpls

Info: Mpls starting, please wait... OK!

[R2-mpls]mpls ldp

[R3]mpls lsr-id 4.4.4.4

[R3]mpls

Info: Mpls starting, please wait... OK!

[R3-mpls]mpls ldp

Configure MPLS and LDP on interfaces of MPLS routers.

[R1]interface Serial 1/0/0

[R1-Serial1/0/0]mpls

[R1-Serial1/0/0]mpls ldp

[R2]interface Serial 1/0/0

[R2-Serial1/0/0]mpls

[R2-Serial1/0/0]mpls ldp

[R2-Serial1/0/0]quit

[R2]interface Serial 2/0/0

[R2-Serial2/0/0]mpls

[R2-Serial2/0/0]mpls ldp

[R3]interface Serial 2/0/0

[R3-Serial2/0/0]mpls

[R3-Serial2/0/0]mpls ldp

After the configurations are complete, run the **display mpls ldp session** command on Routers. You can see that the status of local LDP sessions between R1 and R2 and between R1 and R3 are **Operational**.

[R1]display mpls ldp session

LDP Session(s) in Public Network

Codes: LAM(Label Advertisement Mode), SsnAge Unit(DDDD:HH:MM)

A '*' before a session means the session is being deleted.

```

-----
PeerID           Status      LAM  SsnRole  SsnAge      KASent/Rcv
-----
3.3.3.3:0        Operational DU   Passive 0000:00:10 41/41
-----
TOTAL: 1 session(s) Found.

```

[R2]display mpls ldp session

LDP Session(s) in Public Network

Codes: LAM(Label Advertisement Mode), SsnAge Unit(DDDD:HH:MM)

A '*' before a session means the session is being deleted.

```

-----
PeerID           Status      LAM  SsnRole  SsnAge      KASent/Rcv
-----
2.2.2.2:0        Operational DU   Active 0000:00:11 46/46
4.4.4.4:0        Operational DU   Passive 0000:00:10 43/43
-----

```

TOTAL: 2 session(s) Found.

[R3]display mpls ldp session

LDP Session(s) in Public Network

Codes: LAM(Label Advertisement Mode), SsnAge Unit(DDDD:HH:MM)

A '*' before a session means the session is being deleted.

PeerID	Status	LAM	SsnRole	SsnAge	KASent/Rcv
3.3.3.3:0	Operational DU	Active	0000:00:11	46/46	

TOTAL: 1 session(s) Found.

Step 4 Establish LDP LSPs.

All LSRs are triggered to establish LDP LSPs based on the host route, which is the default trigger policy.

Run the **display mpls ldp lsp** command on LSRs. All host routes are triggered to establish LDP LSPs.

[R1]display mpls ldp lsp

LDP LSP Information

DestAddress/Mask	In/OutLabel	UpstreamPeer	NextHop	OutInterface
2.2.2.2/32	3/NULL	3.3.3.3	127.0.0.1	InLoop0
*2.2.2.2/32	Liberal/1024		DS/3.3.3.3	
3.3.3.3/32	NULL/3	-	10.0.12.2	S1/0/0
3.3.3.3/32	1024/3	3.3.3.3	10.0.12.2	S1/0/0
4.4.4.4/32	NULL/1025	-	10.0.12.2	S1/0/0
4.4.4.4/32	1025/1025	3.3.3.3	10.0.12.2	S1/0/0

TOTAL: 5 Normal LSP(s) Found.

TOTAL: 1 Liberal LSP(s) Found.

TOTAL: 0 Frr LSP(s) Found.

A '*' before an LSP means the LSP is not established

A '*' before a Label means the USCB or DSCB is stale

A '*' before a UpstreamPeer means the session is in GR state

A '*' before a DS means the session is in GR state

A '*' before a NextHop means the LSP is FRR LSP

[R2]display mpls ldp lsp

LDP LSP Information

DestAddress/Mask	In/OutLabel	UpstreamPeer	NextHop	OutInterface
2.2.2.2/32	NULL/3	-	10.0.12.1	S1/0/0
2.2.2.2/32	1024/3	2.2.2.2	10.0.12.1	S1/0/0
2.2.2.2/32	1024/3	4.4.4.4	10.0.12.1	S1/0/0
*2.2.2.2/32	Liberal/1024		DS/4.4.4.4	
3.3.3.3/32	3/NULL	2.2.2.2	127.0.0.1	InLoop0
3.3.3.3/32	3/NULL	4.4.4.4	127.0.0.1	InLoop0
*3.3.3.3/32	Liberal/1024		DS/2.2.2.2	
*3.3.3.3/32	Liberal/1025		DS/4.4.4.4	
4.4.4.4/32	NULL/3	-	10.0.23.3	S2/0/0
4.4.4.4/32	1025/3	2.2.2.2	10.0.23.3	S2/0/0
4.4.4.4/32	1025/3	4.4.4.4	10.0.23.3	S2/0/0
*4.4.4.4/32	Liberal/1025		DS/2.2.2.2	

TOTAL: 8 Normal LSP(s) Found.

TOTAL: 4 Liberal LSP(s) Found.

TOTAL: 0 Frr LSP(s) Found.

A '*' before an LSP means the LSP is not established

A '*' before a Label means the USCB or DSCB is stale

A '*' before a UpstreamPeer means the session is in GR state

A '*' before a DS means the session is in GR state

A '*' before a NextHop means the LSP is FRR LSP

[R3]display mpls ldp lsp

LDP LSP Information

DestAddress/Mask	In/OutLabel	UpstreamPeer	NextHop	OutInterface
2.2.2.2/32	NULL/1024	-	10.0.23.2	S2/0/0
2.2.2.2/32	1024/1024	3.3.3.3	10.0.23.2	S2/0/0
3.3.3.3/32	NULL/3	-	10.0.23.2	S2/0/0
3.3.3.3/32	1025/3	3.3.3.3	10.0.23.2	S2/0/0
4.4.4.4/32	3/NULL	3.3.3.3	127.0.0.1	InLoop0
*4.4.4.4/32	Liberal/1025		DS/3.3.3.3	

TOTAL: 5 Normal LSP(s) Found.

TOTAL: 1 Liberal LSP(s) Found.

TOTAL: 0 Frr LSP(s) Found.

A '*' before an LSP means the LSP is not established

A '*' before a Label means the USCB or DSCB is stale

A '*' before a UpstreamPeer means the session is in GR state

A '*' before a DS means the session is in GR state

A '*' before a NextHop means the LSP is FRR LSP

In most cases, the default trigger policy is used. The establishment of an LDP LSP is triggered in Host mode.

Change the trigger policy to All on LSRs so that all static routes and IGP entries can trigger the establishment of the LDP LSPs.

```
[R1]mpls
```

```
[R1-mpls]lsp-trigger all
```

```
[R2]mpls
```

```
[R2-mpls]lsp-trigger all
```

```
[R3]mpls
```

```
[R3-mpls]lsp-trigger all
```

After the configuration is complete, run the **display mpls ldp lsp** command on each node to view the established LDP LSPs.

```
[R1]display mpls ldp lsp
```

```
LDP LSP Information
```

DestAddress/Mask	In/OutLabel	UpstreamPeer	NextHop	OutInterface
2.2.2.0/24	3/NULL	3.3.3.3	2.2.2.2	Loop0
2.2.2.2/32	3/NULL	3.3.3.3	127.0.0.1	InLoop0
*2.2.2.2/32	Liberal/1024		DS/3.3.3.3	
*3.3.3.0/24	Liberal/3		DS/3.3.3.3	
3.3.3.3/32	NULL/3	-	10.0.12.2	S1/0/0
3.3.3.3/32	1024/3	3.3.3.3	10.0.12.2	S1/0/0
4.4.4.4/32	NULL/1025	-	10.0.12.2	S1/0/0
4.4.4.4/32	1025/1025	3.3.3.3	10.0.12.2	S1/0/0

10.0.1.0/24	3/NULL	3.3.3.3	10.0.1.1	GE0/0/1
*10.0.1.0/24	Liberal/1026		DS/3.3.3.3	
10.0.2.0/24	NULL/1027	-	10.0.12.2	S1/0/0
10.0.2.0/24	1027/1027	3.3.3.3	10.0.12.2	S1/0/0
10.0.12.0/24	3/NULL	3.3.3.3	10.0.12.1	S1/0/0
*10.0.12.0/24	Liberal/3		DS/3.3.3.3	
10.0.23.0/24	NULL/3	-	10.0.12.2	S1/0/0
10.0.23.0/24	1026/3	3.3.3.3	10.0.12.2	S1/0/0

TOTAL: 12 Normal LSP(s) Found.

TOTAL: 4 Liberal LSP(s) Found.

TOTAL: 0 Frr LSP(s) Found.

A '*' before an LSP means the LSP is not established

A '*' before a Label means the USCB or DSCB is stale

A '*' before a UpstreamPeer means the session is in GR state

A '*' before a DS means the session is in GR state

A '*' before a NextHop means the LSP is FRR LSP

[R2]display mpls ldp lsp

LDP LSP Information

DestAddress/Mask	In/OutLabel	UpstreamPeer	NextHop	OutInterface

*2.2.2.0/24	Liberal/3		DS/2.2.2.2	
2.2.2.2/32	NULL/3	-	10.0.12.1	S1/0/0
2.2.2.2/32	1024/3	2.2.2.2	10.0.12.1	S1/0/0
2.2.2.2/32	1024/3	4.4.4.4	10.0.12.1	S1/0/0
*2.2.2.2/32	Liberal/1024		DS/4.4.4.4	
3.3.3.0/24	3/NULL	2.2.2.2	3.3.3.3	Loop0

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3.3.3.0/24	3/NULL	4.4.4.4	3.3.3.3	Loop0
3.3.3.3/32	3/NULL	2.2.2.2	127.0.0.1	InLoop0
3.3.3.3/32	3/NULL	4.4.4.4	127.0.0.1	InLoop0
*3.3.3.3/32	Liberal/1024		DS/2.2.2.2	
*3.3.3.3/32	Liberal/1025		DS/4.4.4.4	
*4.4.4.0/24	Liberal/3		DS/4.4.4.4	
4.4.4.4/32	NULL/3	-	10.0.23.3	S2/0/0
4.4.4.4/32	1025/3	2.2.2.2	10.0.23.3	S2/0/0
4.4.4.4/32	1025/3	4.4.4.4	10.0.23.3	S2/0/0
*4.4.4.4/32	Liberal/1025		DS/2.2.2.2	
10.0.1.0/24	NULL/3	-	10.0.12.1	S1/0/0
10.0.1.0/24	1026/3	2.2.2.2	10.0.12.1	S1/0/0
10.0.1.0/24	1026/3	4.4.4.4	10.0.12.1	S1/0/0
*10.0.1.0/24	Liberal/1026		DS/4.4.4.4	
10.0.2.0/24	NULL/3	-	10.0.23.3	S2/0/0
10.0.2.0/24	1027/3	2.2.2.2	10.0.23.3	S2/0/0
10.0.2.0/24	1027/3	4.4.4.4	10.0.23.3	S2/0/0
*10.0.2.0/24	Liberal/1027		DS/2.2.2.2	
10.0.12.0/24	3/NULL	2.2.2.2	10.0.12.2	S1/0/0
10.0.12.0/24	3/NULL	4.4.4.4	10.0.12.2	S1/0/0
*10.0.12.0/24	Liberal/3		DS/2.2.2.2	
*10.0.12.0/24	Liberal/1027		DS/4.4.4.4	
10.0.23.0/24	3/NULL	2.2.2.2	10.0.23.2	S2/0/0
10.0.23.0/24	3/NULL	4.4.4.4	10.0.23.2	S2/0/0
*10.0.23.0/24	Liberal/1026		DS/2.2.2.2	
*10.0.23.0/24	Liberal/3		DS/4.4.4.4	

TOTAL: 20 Normal LSP(s) Found.

TOTAL: 12 Liberal LSP(s) Found.

TOTAL: 0 Frr LSP(s) Found.

A '*' before an LSP means the LSP is not established

A '*' before a Label means the USCB or DSCB is stale

A '*' before a UpstreamPeer means the session is in GR state

A '*' before a DS means the session is in GR state

A '*' before a NextHop means the LSP is FRR LSP

[R3]display mpls ldp lsp

LDP LSP Information

DestAddress/Mask	In/OutLabel	UpstreamPeer	NextHop	OutInterface
2.2.2.2/32	NULL/1024	-	10.0.23.2	S2/0/0
2.2.2.2/32	1024/1024	3.3.3.3	10.0.23.2	S2/0/0
*3.3.3.0/24	Liberal/3		DS/3.3.3.3	
3.3.3.3/32	NULL/3	-	10.0.23.2	S2/0/0
3.3.3.3/32	1025/3	3.3.3.3	10.0.23.2	S2/0/0
4.4.4.0/24	3/NULL	3.3.3.3	4.4.4.4	Loop0
4.4.4.4/32	3/NULL	3.3.3.3	127.0.0.1	InLoop0
*4.4.4.4/32	Liberal/1025		DS/3.3.3.3	
10.0.1.0/24	NULL/1026	-	10.0.23.2	S2/0/0
10.0.1.0/24	1026/1026	3.3.3.3	10.0.23.2	S2/0/0
10.0.2.0/24	3/NULL	3.3.3.3	10.0.2.1	GE0/0/2
*10.0.2.0/24	Liberal/1027		DS/3.3.3.3	
10.0.12.0/24	NULL/3	-	10.0.23.2	S2/0/0
10.0.12.0/24	1027/3	3.3.3.3	10.0.23.2	S2/0/0
10.0.23.0/24	3/NULL	3.3.3.3	10.0.23.3	S2/0/0
*10.0.23.0/24	Liberal/3		DS/3.3.3.3	

TOTAL: 12 Normal LSP(s) Found.

TOTAL: 4 Liberal LSP(s) Found.

TOTAL: 0 Frr LSP(s) Found.

A '*' before an LSP means the LSP is not established

A '*' before a Label means the USCB or DSCB is stale

A '*' before a UpstreamPeer means the session is in GR state

A '*' before a DS means the session is in GR state

A '*' before a NextHop means the LSP is FRR LSP

Step 5 Configure the LDP inbound policy.

If labels received on R1 are not controlled, R1 will establish a large number of LSPs, consuming large memory.

After an inbound LDP policy is configured, R1 receives label mapping messages only from R2 and establishes LSPs to R2, saving resources.

Run the **display mpls lsp** command on R1. Information about established LSPs is displayed.

[R1]display mpls lsp

LSP Information: LDP LSP

FEC	In/Out Label	In/Out IF	Vrf Name
3.3.3.3/32	NULL/3	-/S1/0/0	
3.3.3.3/32	1024/3	-/S1/0/0	
2.2.2.2/32	3/NULL	-/	
4.4.4.4/32	NULL/1025	-/S1/0/0	
4.4.4.4/32	1025/1025	-/S1/0/0	
10.0.12.0/24	3/NULL	-/	
10.0.1.0/24	3/NULL	-/	
2.2.2.0/24	3/NULL	-/	
10.0.23.0/24	NULL/3	-/S1/0/0	

10.0.23.0/24	1026/3	-/S1/0/0
10.0.2.0/24	NULL/1027	-/S1/0/0
10.0.2.0/24	1027/1027	-/S1/0/0

You can see that LSPs to R2 and R3 are established on R1. Configure the inbound policy on R1 to allow only the routes to R2.

```
[R1]ip ip-prefix prefix1 permit 10.0.12.0 24
```

```
[R1]mpls ldp
```

```
[R1-mpls-ldp]inbound peer 3.3.3.3 fec ip-prefix prefix1
```

```
[R1-mpls-ldp]quit
```

```
[R1]display mpls lsp
```

LSP Information: LDP LSP

FEC	In/Out Label	In/Out IF	Vrf Name
2.2.2.2/32	3/NULL	-/-	
10.0.12.0/24	3/NULL	-/-	
10.0.1.0/24	3/NULL	-/-	
2.2.2.0/24	3/NULL	-/-	

----End

Additional Exercise: Analysis and Verification

How can you configure R1 to receive Label Mapping messages from R1 to R3?

Device Configuration

```
<S1>display current-configuration
```

```
!Software Version V200R008C00SPC500
```

```
#
```